ELSEVIER

Forest Policy and Economics 2 (2001) 101–116

Forest Policy and Economics

www.elsevier.nl/locate/forpol

Increasing rotation periods during a time of decreasing profitability of forestry — a paradox?

Martin Moog*, Herbert Borchert

Received 1 February 2000; received in revised form 9 August 2000; accepted 22 August 2000

Abstract

The profitability of forestry in Germany decreased during recent decades because of more or less constant prices for forest products, increasing input prices and limited success in rationalisation. However, the volume of growing stock increased significantly during this period since forest enterprises have chosen longer rotation periods. This is especially true for state owned and community owned forests but also at least partly for private forests. It is normally not an alternative for the owners of forest enterprises in Germany to sell the forests completely, but on a first glance increasing investments in standing timber during a time of decreasing profitability of forestry seems to be inconsistent with economic theory. On the one hand this observation could be explained as the result of non-timber values. However, this paper is focused on another approach, which is an expanded Faustmann model in line with soil rent theory and focused on timber production. Profitable rotations in the future have the effect of shortening the optimal rotation period because an investment in standing timber causes opportunity costs by delaying the establishment of the next generation of the forest. Unprofitable future rotations have the opposite effect, if the landowner is forced to reforest. In case investments in reforestations are not profitable decision-makers have good reasons not to cut the mature stands, in spite of the fact that the internal rates of return of investments in standing timber are low in comparison with investments on the financial market. Empirical data for the period 1954–1998 mostly from guidelines for forest valuation are used together with inflation corrected interest rates to show that optimal rotation length increased over time. Nevertheless we have to recognise that the observed rotation periods are distinctly longer than the calculated optimal rotation lengths. Other factors which may also explain the investments in forestry are discussed later. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Optimal rotation period; Faustmann model; Presslers indicating percent; Profitability of forestry; Germany

* Corresponding author. Tel.: +49-8161-71-4630; fax: +49-8161-71-4631.
E-mail address: moog@forst.uni-muenchen.de (M. Moog).

PII: S1389-9341(01)00047-8

1. Economic development of German forestry during recent decades

During the past decades the economic situation of forest enterprises in Germany has deteriorated. An empirical study by Brabänder and Behrndt (1982) shows drastically how the forestry situation deteriorated between 1955 and 1980. This decline of profitability continues to the present time. The main reason for this development, is the change in relations between product prices and the prices of production factors, especially labour costs. Timber prices did not show an enduring increase while wages increased constantly at a high rate.

The price–cost gap forced German forestry to rationalise forest management and considerable successes were reached in this field. Productivity increased considerably after the introduction and improvement of the chainsaw and later by introducing modern harvesting machinery. A calamitous storm in central Europe in 1990 provided the opportunity of establishing modern harvester technology in Germany. Only with the aid of this technology could forestry cope with the damage caused by the gale. However, the rationalisation efforts could not prevent forestry from becoming more and more unprofitable. Wood will possibly become a more important raw material in the future than it is at present, but the prospects for German forestry were bad during the past decades and are still the same now.

2. Investments in German forestry

Considering the development of the economic situation a reduction in investments in forestry must be expected, because from the point of view of economists it is certainly unwise to invest in an unprofitable economic sector. However, contrary to this expectation forest enterprises in Germany increased investments in forestry during past decades. Some evidence for the remarkable investments in forest that were made during the time period in question must be given. Considering investments in Germany as one economic area purchases of forests are not an important aspect. Moreover, there is actually no active trade with forestland in Germany. Forestation of agricultural land took place to a certain degree, but there was only a marginal increase of forestland during the past decades. Under German conditions investments in forestation of agricultural land are normally not the result of a calculation in which profitability of forests is of importance. Forestation is either the result of a lack of profitability of agriculture or the result of special subsidies given to farmers to reduce the production of agricultural products. In the first case natural succession often takes place. Additionally it is not easy to get the necessary permission for forestations.

The only important possibility of investing in forestry is to increase the standing timber of old forests (Kroth, 1968, p. 178). Obviously, German forest enterprises invested considerably in growing stock of mature forests during the past 50 years. During the 20th century the growing stock probably reached its minimum some years after the end of World War II. This is due to the high harvest rates during the war and the harvest necessary to satisfy claims of the allies and vital essentials of the German population after the war. Since 1950 the growing stock in German forests increased continuously reaching a very high level in recent years.

Evidence for this is given in the yearly reports of state forest administrations and by the results of forest inventories. However, data of only one nation-wide forest inventory are available. These data are from 1987 and can be compared to a forest inventory of Bavarian forests from 1971. In the 1987 inventory no significant difference between private forests and state forests regarding standing volume of timber could be found. So the assumption may be allowed that private forest owners behaved similarly to the state forest management regarding investments in standing timber.

Sadly data about standing timber are not available in the yearly reports before the 1970s and the quality of the data differs much between state forest administrations. Comparatively good information is available about the volume of standing timber in the state forests of Bavaria. In the

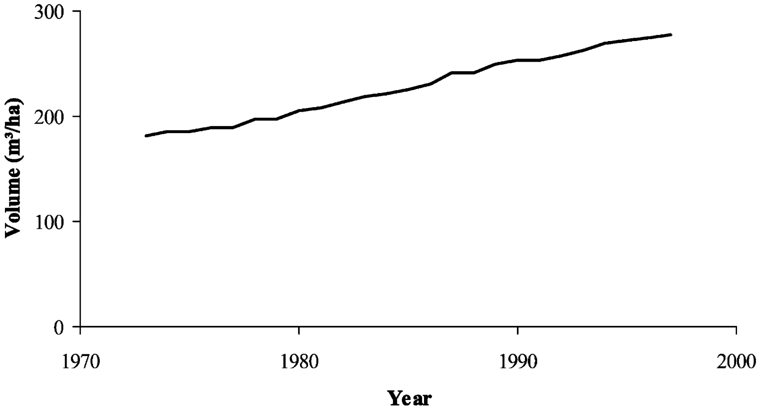

Fig. 1. Development of the growing stock in the Bavarian State Forests between 1973 and 1997. The forest area decreased during this time from approximately 0.75 to 0.72 million hectares and the proportion of coniferous from 76% to 72%.

Bavarian State forests the volume of standing timber per hectare has increased by 93 m^3 or approximately 4 m^3/year and hectare from 1973 to 1997 (Fig. 1). Because of different time periods a comparison should be made using the average increase in percent per year calculated with the formula for compound interest. Standing timber increased in the different state forests between 0.3 and 1.8% per year.

Since most German forests are managed following the model of an age class forest with the rotation period as the most important ratio of forest management it may be assumed that the average age of the forests which are harvested increased during recent decades and also the targets concerning rotation periods have changed. However, data about the average age of the forests which are harvested are not published and it is also difficult to find information about the rotation periods used in forest management plans during the time period in question. Instead of using data on the age of stands, final cut rotation periods were calculated from the proportion of the age classes recorded by forest inventories (Bundesministerium für Ernahrung, 1953; Kaiserliches Statistisches Amt, 1903, 1916; Statistisches Reichsamt, 1943, 1930; Statistisches Bundesamt Wiesbaden, 1966).

Fig. 2 shows the development of the average rotation periods of beech, spruce and pine in the Bavarian State forests between 1900 and 1997. For spruce and pine a noticeable increase of the rotation periods in the last four decades is shown. However, the rise is lower than expected from the data of the development of the growing stock in Fig. 1. The reason might be the increase in current annual growth of timber in the last decades that have been determined for forests in Germany. The current annual growth of timber of stands at the same age in Bavarian forests increased for instance between 1971 and 1987, approximately one-third for spruce, over 40% for pine and approximately 10% for beech (Pretzsch, 1999, S. 241). Similar changes in forest growth were observed in other parts of Europe (Spiecker et al., 1996). Increased growth rates can be found for all age classes of forests. Since there is no significant change in the form of the curve of growth as a function of age, a change in optimal rotations must not be expected. Moreover, the decision-makers in German forestry were not aware of the actual development. Up to recently many decision-makers were even convinced that growth was depressed because of environmental impacts. Thus it is very unlikely that increased growth rates could have influenced forest management before the 1990s. Thus, increased growth may be one reason for the actual increase in standing timber which was partly not intended by foresters and forest owners. For beech forests Fig. 2 is misleading, since data were calculated from data of the age class distribution. Sadly the number of age classes was reduced over time to

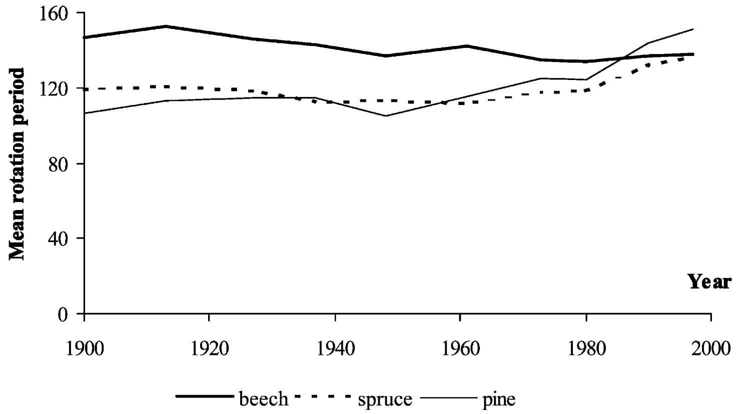

Fig. 2. Development of the mean rotation period of stands in the Bavarian state forests.

only seven 20-year age classes for the present time; in contrast to up to 10 age classes were used before. In the calculation it was assumed that the oldest age class comprised a period of 20 years like the other age classes while actually many forests are older. Thus, the rotation periods are underestimated, especially since 1970. Particularly the rotation period of stands of beech is today certainly longer. The oldest age class comprised for beech in 1997 is approximately 60% more than the average area of the other classes.

3. Decreasing profitability explains increasing rotation periods

Having enough empirical evidence to assume that forest owners invested considerably in standing timber by increasing rotation periods during recent decades the question is how this can be explained especially on the basis of the soil rent theory using the Faustmann formula.

A common use of the Faustmann formula is to calculate optimal rotation periods when data regarding regeneration costs, amounts of timber, timber prices and harvest costs are known and an interest rate is given.

It can easily be shown that using the Faustmann formula the result for the optimal rotation period differs from the result which can be gained by a calculation for only one rotation using the NPV (Kula 1988; Neher, 1990). This difference is due to the opportunity costs of the following rotations. In the case of a profitable forest and a positive land expectation value the result gained for optimal rotation with the Faustmann approach is a shorter time period than that gained by using the NPV formula for a single rotation. This can clearly be seen when we look at the formula known as Pressler's indicating percent (PI-%).

$$\text{PI-\%} = \frac{\text{current annual value growth}}{\text{stumpage value} + \text{soil expectation value} - \text{capitalised costs for administration}} \tag{1}$$

Eq. (1) is a simplified version of a formula first obtained by Pressler (1860). Pressler's indicating percent is the marginal rate of return of a mature forest. It is the ratio between the expected increase in net value of the forest during the coming year and the sum of its net value at the beginning of the year and the land expectation value minus the capitalised costs for administration. The numerator is the yield and the denominator is the input in terms of opportunity costs. Leaving the forest for another year means to do without the net value of the forest and not to replant the forestland. The land expectation value stands for the profit which will be gained by replanting the land. The capitalised costs of administration are not relevant for the decision and are assumed to be constant in this model. Therefore this term is added to the denominator to equalise the same term being the second term in the Faustmann formula.

The indicating percent is the ratio used to decide whether or not a mature forest should be cut and the land replanted. It must therefore be compared with the expected rate of return and forests with indicating percents below this rate should be harvested. It can be assumed that the indicating percent is a good representation of the calculation done by forest owners when deciding about investments or disinvestments in standing timber. Now analysing the effect of a decreasing (future) profitability of forestry we see that a lower land expectation value c.p. leads to a higher indicating percent.

While in nearly all publications positive land expectation values are assumed it is a common conviction that the more correct calculation of the optimal rotation period using the Faustmann formula leads to shorter rotation periods than the more simple NPV approach for a single rotation. However, in the case of negative land expectation values the Faustmann approach leads to longer rotations.

A negative land expectation value indicates that forestation or reforestation are not profitable investments and alternative investments should be chosen instead. For the owner of a mature forest this means that he should not replant the land after harvesting the forest. If there are no other costs remaining after harvesting the growing stock (i.e. taxes for landowners) a forest owner can calculate the optimal time to harvest his forest using Pressler's indicating percent with a land expectation value equal to zero. In this case Pressler's indicating percent is equal to the proportional rate of valuegrowth of the stand (Wertzuwachsprozent) and the solution gained using the indicating percent is equal the solution for a single rotation. However, forest owners are forced by German forest legislation to replant their forests and to maintain young forests to a certain degree. This means that they are forced to make unprofitable investments under certain conditions. This is the reason why it would make sense for German forest owners to use Pressler's indicating percent to calculate optimal time of harvests using negative values for the land expectation value in the calculation.

Naturally the question occurs if this effect can be used to explain the decisions of German forest owners to invest in standing timber instead of cutting forests and replanting the land.

4. Data sources

To calculate a series of representative land expectation values for the most important tree species in Germany data published by the state forest administration of Nordrhein-Westfalen for forest valuations (Ministerium für Umwelt, 1954–2000 Raumordnung und Landwirtschaft) can be used. The tree species are Norway spruce

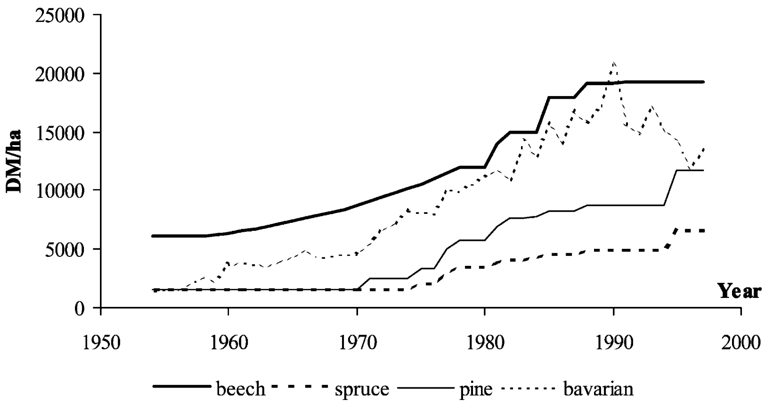

Fig. 3. Data to regeneration costs for different tree species in Nordrhein-Westfalen and average costs in Bavaria.

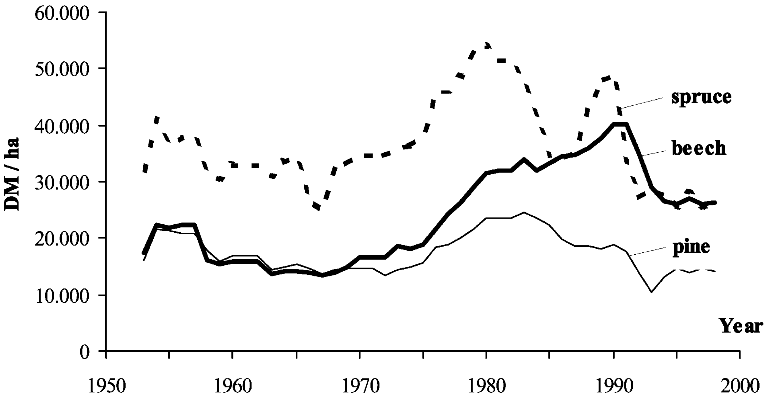

Fig. 4. Stumpage value of stands of spruce, beech and pine on an average site class at ages of 80, 120 and 120 years, respectively.

(*Picea abies*), pine (*Pinus sylvestris*) and beech (*Fagus sylvatica*). Time series of these data are shown in Figs. 3 and 4. Regeneration costs show a remarkable increase over time and differ much between tree species. They are significantly lower for Norway spruce than for pine and beech. Average reforestation costs reported by the state forest of Bavaria are presented in Fig. 3 for comparison.

Under German conditions artificial regeneration of beech and other broadleaf forests are much more expensive than regeneration of coniferous forests, however, Norway spruce can be planted much cheaper than pine. Data used are shown in Fig. 3. Since the source of data are guidelines for the valuation of forests there are steps in the time series which are not caused by drastic increases of costs in the particular year (i.e. 1995) but because price alignment was not carried out currently. However, the time series for regeneration costs were not smoothed for the calculations presented.

Stumpage values of stands on average site classes and for common ages of final cutting are shown in Fig. 4. Stumpage values of mature stands of pine and spruce are at present not higher than in 1955. Both show a peak at the beginning of the 1980s. For beech stands a significant increase of stumpage value during the time period in question can be observed with a slightly decreasing price in the period before 1968 and a continuous increase from 1968 to 1992. Because the prices used by the publishers of the forest valuation guidelines are not prices of the particular year but average prices over a short period the time series of stumpage value of mature stands are smoother than time series of timber prices.

For the three tree species the age intervals with available data on stumpage value were limited and changed twice as noted in Table 1.

For the first two periods data on stumpage value were available for steps of 10 years, for the last period for annual steps. For the calculation of optimal rotation periods these discrete data were transformed in continuous data by computing curves using regression. These curves show stumpage value as a function of the age of the stands.

Table 1
The age intervals with available data on stumpage value

Tree species	Available data for the ages in time period		
	1954–1978	1979–1987	1988–1997
Spruce	60–100	40–120	60–100
Pine	30–140	40–140	80–120
Beech	50–140	80–140	100–140

The results regarding the optimal rotation period must be evaluated carefully in all cases where the calculated rotation period is outside of the age intervals given in Table 1.

Normally thinnings should be taken into account when optimal rotation periods are calculated with the Faustmann formula. However, it is difficult to get data for the net values of thinnings which are mostly marginal. To choose a solution without ignoring thinnings while restricting the complexity of calculations the stumpage values at the age of rotation were multiplied by the percentage of timber harvested before the final cut (data from yield tables) when land expectation values are calculated.

Choice of the right interest rate for the calculation is the most difficult problem in this empirical study. This problem is solved by using the annual average rate of return of long running securities, corrected by the rate of inflation to get an inflation free interest rate (Fig. 5).

This interest rate is well below 5% during most years with an average of 4.2%. Periods with high interest rates were the years 1967–1970 and 1986–1991.

5. Land expectation values under German conditions during recent decades

The calculations with the Faustmann formula (without capitalised costs of administration) using rotation periods of 80 years for Norway spruce, 100 years for pine and 120 years for beech leads to the results shown in Fig. 6. The results are negative for pine forests and for beech forests during the whole time period in question. Positive values are calculated for spruce forests up to 1980. However, since that time also the soil expectation values for spruce forests are also negative. If capitalised costs of administration were subtracted the values would be even lower. Land expectation values calculated with a constant interest rate of 3% do not differ much from this result for beech and pine (Fig. 7). For spruce in this case the land expectation value is positive until 1987. In comparison with Fig. 6 the run of the curves in case of a constant interest rate (Fig. 7) are smoother.

6. Economic impacts on the optimal rotation length

Substituting the interest rate r for Pressler's indicating percent we can derive from Eq. (1)

$$\begin{aligned}\text{current annual value growth} &= r\cdot\text{stumpage value}\\ &\quad + r\cdot\text{soil expectation value}\end{aligned} \tag{2}$$

while assuming the costs of administration are not included in the soil expectation value. Eq. (2)

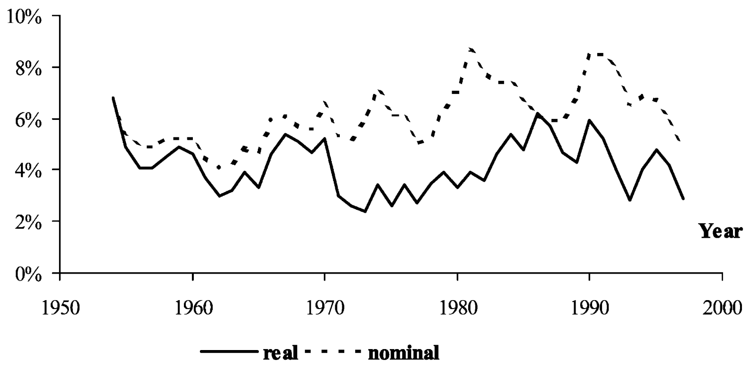

Fig. 5. Average rate of return of long running securities in Germany.

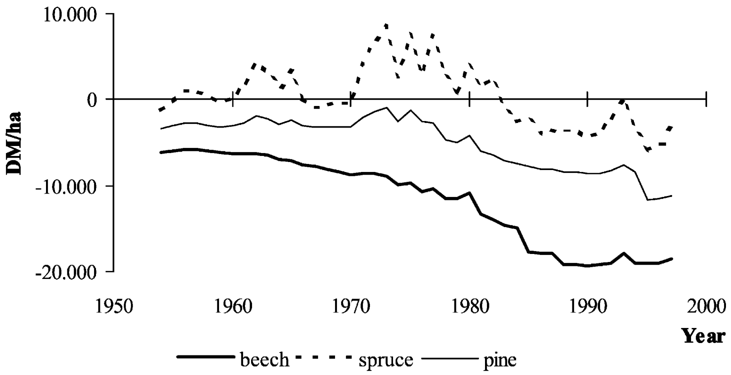

Fig. 6. Soil expectation values for rotation periods of 80 years for spruce, 100 for pine and 120 for beech at average site classes calculated with the real rate of return of long running securities.

is the first order condition of the maximum of the soil expectation value with respect to rotation length. The first term on the right side of Eq. (2) indicates the interest costs and the second term represents the soil rent. Both are opportunity costs and account for the capital charges. The optimal rotation period is indicated if the current annual value growth and the capital charge amounts are equal (Fig. 8).

In case of a negative soil expectation value the soil rent is also negative. If reforestation is statutory we have to consider the negative soil rent in calculating the capital charges. Thus the level of the capital charges is below the level of the interest rate as shown in Fig. 9.

On the basis of Figs. 8 and 9 we can regard the impact of different parameters. Increasing regeneration costs only affect the soil rent by reducing this. Consequently the capital charges decrease and the rotation length should be extended. Increasing interest rates raise the interest costs and the soil rent if the latter has a positive value, otherwise the soil rent will be reduced. Nevertheless, the capital charges increase because the impact on the interest rate is greater. An increasing stumpage value raises both the interest costs and the soil rent. However, the impact on the optimal rotation length is uncertain because the current annual value growth might also be affected. If we assume that the increase strikes all

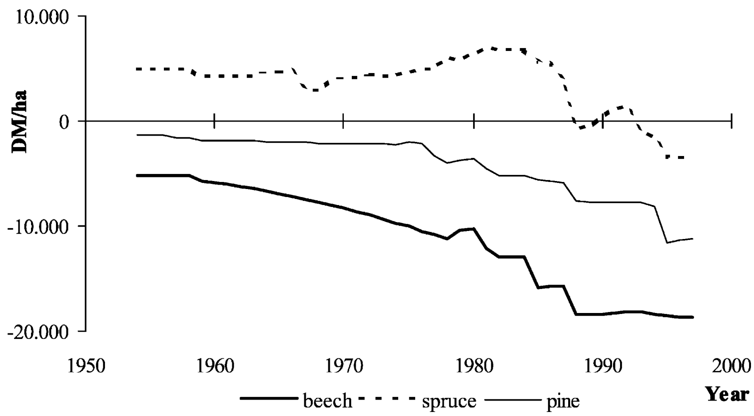

Fig. 7. Soil expectation values for rotation periods of 80 years for spruce, 100 for pine and 120 for beech at medium site classes calculated with a constant interest rate of 3%.

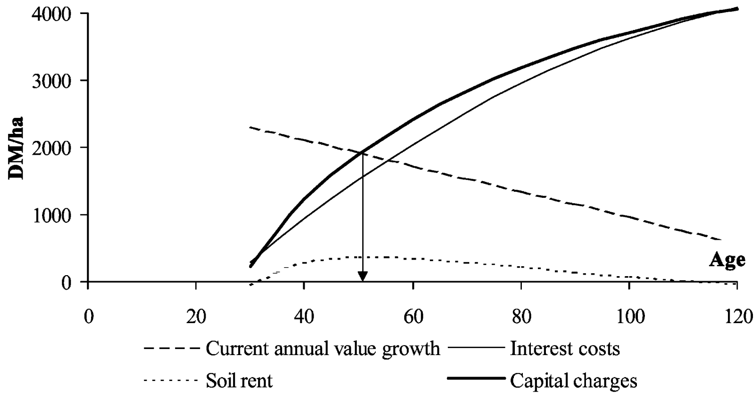

Fig. 8. The development of the current annual value growth, interest costs, soil rent and capital charges over age.

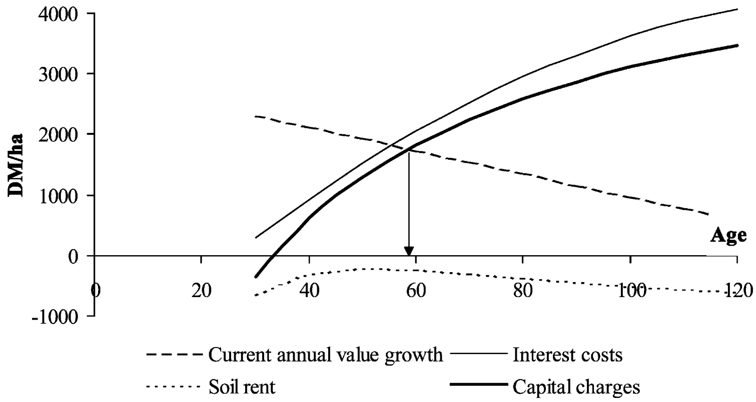

Fig. 9. The development of the current annual value growth, interest costs, soil rent and capital charges over age in case of a negative soil expectation value.

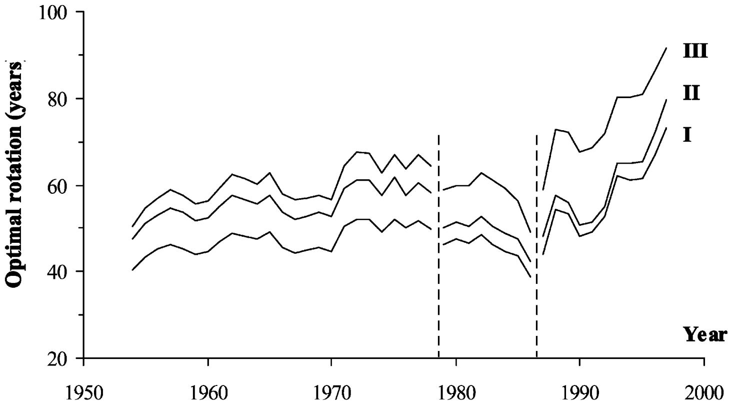

Fig. 10. Optimal rotation of Norway spruce at different site classes. Increasing number of site class indicates a decreasing productivity.

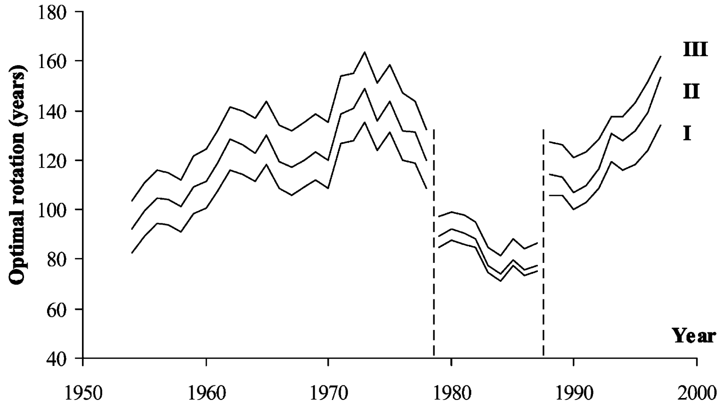

Fig. 11. Optimal rotation of beech at different site classes. Increasing number of site class indicates decreasing productivity.

types of timber likewise the current annual value growth will not change. Thus the rotation length has to be shortened.

7. The development of optimal rotation periods

The optimal rotation periods were calculated using Pressler's indicating percent. Not surprisingly for the higher yield classes shorter optimal rotations are found. Inconsistencies of the time series of data in 1979 and 1987 must be taken into account for interpretation of results. These inconsistencies are due to a change of the model used to structure the amount of timber into different types sold on the market (classes of stems, pulpwood, etc.). Interpretation of the results are therefore made for the time periods marked in Figs. 10–12.

The short-run fluctuations of optimal rotation were very much determined by changes of the rate of interest, whereas the impact of price and cost changes are rather detectable in the long-run trend. The interest rate also controls the amplitude of fluctuations. This can be seen by comparing Figs. 10–12 with Fig. 13. Since stumpage values were used for calculating rotation length the influence of changes in prices and harvesting costs must be analysed separately by using other sources for data. The development of harvesting costs are therefore shown using the piecework rate (Fig. 14) and the (gross-) prices for different types of sawtimber (Fig. 15). These time series data are from the Bavarian state forests. The data

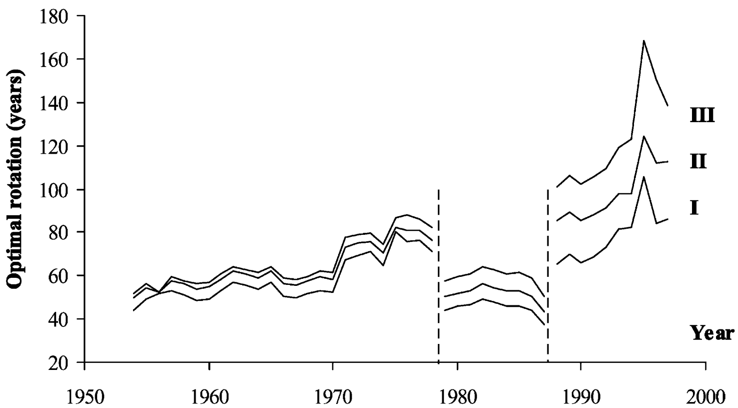

Fig. 12. Optimal rotation of pine at different site classes. Increasing number of site class indicates decreasing productivity.

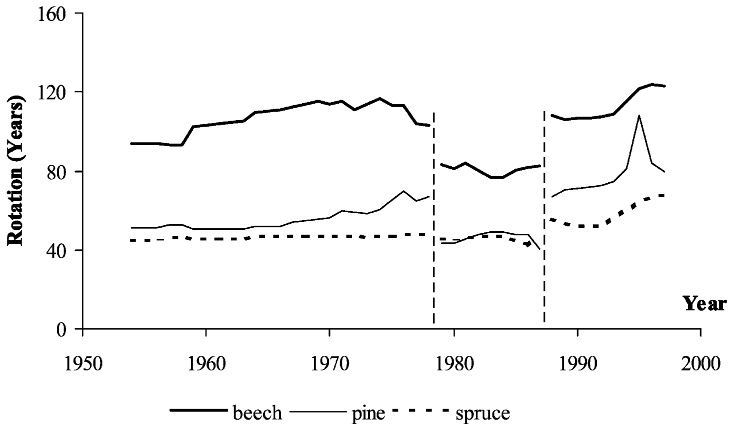

Fig. 13. Optimal rotation on good sites calculated with a mean interest rate of 4.2%.

to regeneration costs of Fig. 3 are shown again in Fig. 16, but now in real factor prices.

From 1954 up to around 1965 we find an increase in optimal rotations which is most significant for beech stands (Fig. 11) and mostly due to the decrease of timber prices during that period. During the following years up to 1970 an increase in the rate of interest nearly compensates the further decline of timber prices resulting in more or less constant optimal rotations. Lower interest rates during the following years until 1973 cause another increase in optimal rotations. From 1976 on we find a decrease of optimal rotations for beech due to an increase of timber prices. The optimal rotation periods for spruce stands (Fig. 10) remain more or less constant during the years up to the end of the first time period, 1978. The changes are determined by changes of the interest rate. A slight increase in costs for regeneration at the end of the first period (Fig. 16) was compensated by a slight increase in timber prices. For pine forests we find increasing optimal rotations due to increasing regeneration costs (Fig. 16) and continuously decreasing prices (Fig. 15).

During the time period from 1979 to 1987 we generally find decreasing optimal rotations which is due to increasing rates on the market for capital. Only the rotation for spruce was noticeably raised at the end of this period.

The changes at the beginning of the third time period (1988) are mainly due to changes in the interest rate. After 1991 there is a significant

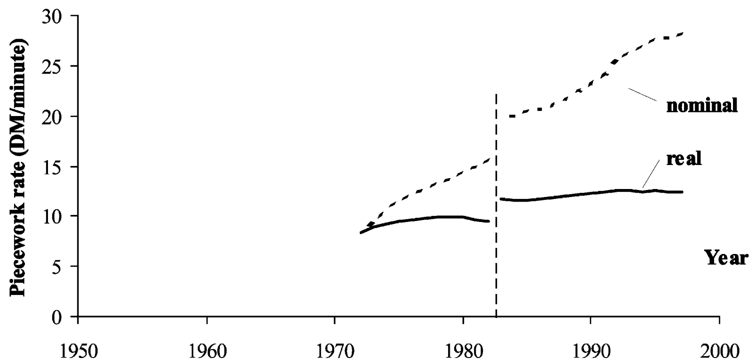

Fig. 14. Piecework rate for harvesting in state owned forests. The jump from 1982 to 1983 is due to the implementation of a new collective labour agreement. It is not the result of a wage rise. In fact the reference of the piecework rate changed.

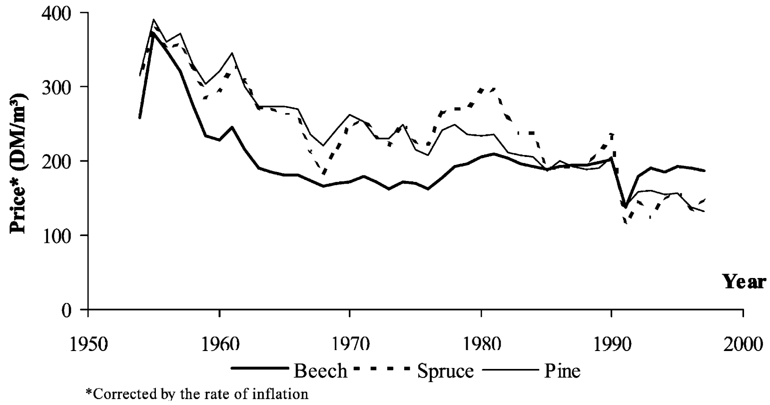

Fig. 15. Inflation corrected prices for certain grades of sawtimber of beech, spruce and pine calculated from data of Bavarian state forests.

increase of the optimal rotation for all tree species. The reasons for this were storm damage in 1990. This caused an enormous increase in timber supply resulting in sharply decreasing prices for all sorts of timber with further price drops caused by barkbeetle calamities. Together with decreasing rates of interest this caused a significant increase in optimal rotations. Most significant is a sharp increase in the rotation periods for pine forests which is due to an increase of planting costs for this species.

8. Discussion

Decreasing profitability of forestry and increasing volume of standing timber are not contradictory. At first glance it naturally seems to be irrational since decreasing investments should be expected when the economic expectations regarding an economic sector become worse. However, both are in line with the land rent theory if reforestation is statutory but not profitable. Thus there exists no paradox. Nevertheless we have to recognise that the observed rotation periods (Fig. 2) are far beyond the calculated optimal rotation lengths (Figs. 10–12). The question that matters most is whether there are other reasons than the relative increase in the marginal rate of interest of forests that can explain the decisions of forest owners.

There are at least five other approaches or objections against the explanation presented in this paper that should be mentioned here.

The first objection is that the calculations pre-

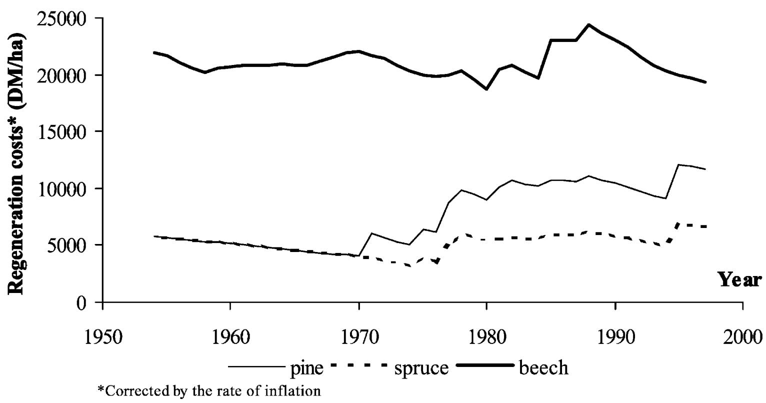

Fig. 16. Regeneration costs corrected by the rate of inflation.

sented in this paper are not done with the relevant data. Another explanation for the observed increase of investments is that profits from forestry are less important than the non-market products and therefore decisions about rotation periods are more influenced by non-market benefits. A further explanation is that decision making by German forest owners is generally not done on the basis of models of land rent theory but with the aim to maximise forest rent. Not contrary to this explanation is an approach using the agency theory. This leads to the assumption that the managers of forest enterprises misled the forest owners. At last a silvicultural aspect is discussed regarding its contribution to explain the behaviour of German foresters.

Of course we do not know data assumed by forest owners when making their harvesting and replanting decisions. There is certainly a tradition in German forestry to use actual data for forest valuation but there is no strong evidence that actual data are used which means the average price and cost data from several recent years are used. On the other hand, there is a tradition of optimism in German forestry regarding future prospects and especially the demand of timber in the far future. If the decision-makers in forestry always expected a much better economic situation in the far future the investments in forestry must be believed to be good investments with high rates of return. There is not much evidence for this hypothesis. On the other hand, there is not much evidence to defeat this theory.

Since recreation and protection effects of forests are very much considered in German forestry there is evidence for the assumption that decisions about rotation periods are much more influenced by the non-market products of forests than by calculations about profits from timber production. This hypothesis includes the assumption that forests managed with longer rotation periods generally produce more or better non-market products and long rotations are therefore preferable. This is certainly in line with a widespread opinion in German forestry. Since state forests were managed without significant profits, or even with deficits during many years especially during the last two decades, this hypothesis is certainly not unplausible for the forests which are state owned or owned by communities. Actually, small profits and deficits of state forests were often justified by the managers by emphasising the non-market values of the forests. However, it can be doubted that non-market values play an important role for decision making in private forestry. So at least for a significant share of German forests non-market products are certainly not responsible for the increase in growing stock.

Another aspect related to the theory presented above is that the soil rent theory which is behind the calculations is not very popular in German

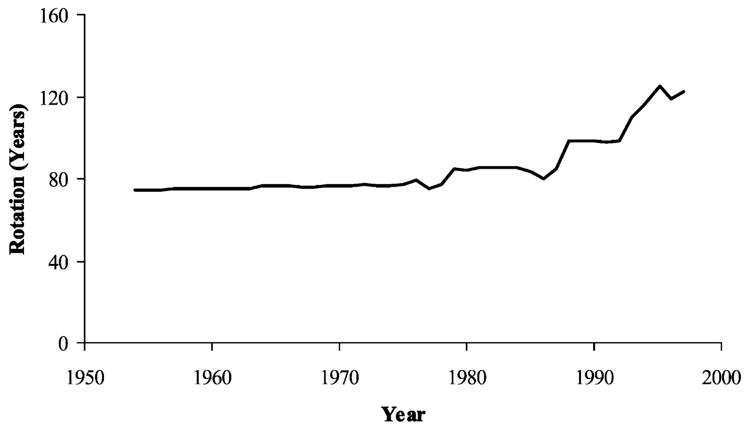

Fig. 17. Rotations for stands of spruce on good sites maximising the forest rent.

forestry. Instead the forest rent theory was, and still is, popular and it can be said that the increase in growing stocks may be the result of maximum forest rent calculations. Much longer rotation periods must be chosen to maximise the cash flow from a forest with no regard to capital costs. Fig. 17 shows rotations for stands of spruce on good sites that yield the highest net revenue. These rotation periods are equal to the rotation periods that could be found in the practice of public owned forests in Germany.

Some publications from managers of state forests could be recorded during past decades comparing forest rents for different rotations (e.g. Ripken and Spellmann, 1980 and Ripken, 1989 for Lower Saxony). In the case of the Bavarian state forest official statements claim that the choice of rotation period should be carried out with the objective of maximising the forest rent (Bayerische Staatsministerium für Ernährung, 1982, p. 20; Schreyer, 1986). A leading manager of the administration of the state forests of Baden-Wuerttemberg stated in 1988 that the rotation periods had been raised in the previous years in order to maximise the growth value and with respect to the recreation and protection effects of forests (Weidenbach, 1988). Hence there is good reason to assume that maximising the forest rent is the reason for choosing longer rotation periods in public owned forests.

While capital bound may be of no relevance for managers of state forests it is hard to believe that capital costs are totally ignored by the owners of private forests. Owners of private forests mostly have alternative investments and should be aware of their opportunity costs. So they hardly decide about rotation periods on the basis of simple calculations following the forest rent theory. On the other hand, it must be realised that management of the state forests has a strong influence on private forests, at least on farm forestry and other small-scale forestry. Also the fact must be considered that the managers of the bigger private forest estates are trained and educated in the same schools being under a strong influence of state forests. The result is that the managers of private forests share values and ideologies with their counterparts from state forests. Drawing into consideration that the forest rent theory is a very typical ideology of German forestry and calculations using compound interest are taboo it is possible that not only state forest managers but also managers of private forests base their decisions on calculations following the forest rent theory. Possibly some of the managers of private forest estates do not want to break this taboo and in teaching forest economics not much emphasis was laid on training the use of models based on soil rent theory during the last two-thirds of the 20th century.

One aspect that may play a role is that the managers of state owned and privately owned forests are agents of the owners. Institutional economics analyses the relation between principals and agents, i.e. the owners of firms and the managers (Kieser, 1995; Picot et al., 1997). It is very common to assume that there is asymmetric information which is used by managers to maximise their utility. To prevent actions of the management which are not in line with the interest of the owners incentives are suggested (i.e. stock options). We do not see any reason why foresters should be better in this respect than managers of industrial firms. Generally there are no incentives for the management of forests to manage the forests with much attention on profitability. There is almost no vertical integration of the forestry and wood processing industry in Germany. Since alternative investments within the forest estate are in most cases not available it seems plausible that managers of forests invest in growing stock when there is money to invest. Possibly the management of the forests may be interested in investments and use misleading information to prevent owners from decisions about harvesting with consideration of capital costs. Since the prestige of the managers is at least partly connected with growing stock and rotation periods there is certainly an incentive to choose long rotation periods and to increase growing stock (Hiley, 1967, p. 294).

A final partly silvicultural aspect is based on the fact that in many cases reforestation is possible not only by planting trees but also by natural regeneration and that the latter is often much cheaper. If the rotation period is long and the rate of interest used is significantly above zero regeneration costs are most important regarding the amount of the land expectation value. Assuming the land expectation value is negative; if a forest owner in case of the final cut of a stand is forced to care for regeneration, lower costs of natural regeneration could make the enforced investment less unprofitable or even move it to success. The forest owner may hope that there will be a natural regeneration in the future and therefore postpone the final harvest. After the proportional rate of growth of the stumpage value (Wertzuwachsprozent) has fallen below the rate of interest there are costs for waiting further. These opportunity costs amount to the difference between the further increase in stumpage value and the earnings of alternative assets. The expected amounts must be compared with the savings from postponement of the harvest regarding regeneration costs. The more significant the difference in costs of replanting and natural regeneration is the longer it is worthwhile to wait.

It is certainly not easy to draw a final conclusion. Even though the behaviour of German foresters regarding public owned and private owned forests seems equal there might be different motives. The fact that reforestation is statutory might play a role in both groups. We presume that the staff of public forest enterprises take none market values to a greater extent into account. However, we think that the institutional conditions can provide the substantial explanations. Therefore we have to treat the different groups of foresters separately. Maybe the approaches of institutional economics can offer a more satisfactory explanation for the observed discrepancy between profitability and capital expenditure in German forestry.

References

Bayerisches Staatsministerium für Ernährung, 1982. Landwirtschaft und Forsten: Forsteinrichtungsrichtlinien.

Brabänder, H.D., Behrndt, M., 1982. Die nominale und reale Entwicklung von Arbeitsaufwand und Holzertrag im langfristigen Zeitvergleich (1955 bis 1980). Der Forst- und Holzwirt 37 (23), S.581–S.587.

Bundesministerium für Ernährung, 1953. Landwirtschaft und Forsten (Hrsg.): Die Forsterhebung 1948. Bonn.

Hiley, W.E., 1967. Woodland Management, 2nd ed. Faber and Faber Limited, London.

Kaiserliches Statistisches Amt (Hrsg.), 1903. Die Forsten und Holzungen im Deutschen Reich nach der Erhebung des Jahres 1900. In: Vierteljahreshefte zur Statistik des Deutschen Reichs. Ergänzungsheft zu 1903, II. Berlin: Verlag von Puttkammer u. Mühlbrecht.

Kaiserliches Statistisches Amt (Hrsg.), 1916. Die Forsten und Holzungen im Deutschen Reiche nach der Erhebung des Jahres 1913. In: Vierteljahreshefte zur Statistik des Deutschen Reichs. Heft 4. Berlin: Verlag von Puttkammer und Mühlbrecht.

Kieser, A. (Hrsg.), 1995. Organisationstheorien. 2. Auflage. Stuttgart (u.a.): Kohlhammer Verlag.

Kroth, W., 1968. Der forstliche Produktionszeitraum. Möglichkeiten optimaler Betriebsgestaltung in der Forstwirtschaft, Bayerischer Landwirtschaftsverlag, München, S.173–184.

Kula, E., 1988. The Economics of Forestry. Modern Theory and Practice. Timber Press, Portland, pp. S138–S146.

Ministerium für Umwelt, 1954–2000. Raumordnung und Landwirtschaft Nordrhein-Westfalen: Richtlinien zur Waldbewertung in Nordrhein-Westfalen (WBR).

Neher, Ph.A., 1990. Natural Resource Economics. Conservation and Exploitation. Cambridge University Press, Cambridge, pp. S62–S80.

Picot, A., Dietl, H., Franck, E., 1997. Organisation. Eine ökonomische Perspektive. Schäffer-Poeschel Verlag, Stuttgart.

Pressler, M.R., 1860. Zur Verständigung über den Reinertragswaldbau und dessen Betriebsideal. Allgemeine Forst- und Jagd-Zeitung 1860. 1.Artikel S. 6–55 und 2. Artikel S. 173–191.

Pretzsch, H., 1999. Waldwachstum im Wandel. Forstwissenschaftliches Centralblatt Jg. 118 (4), 228–250.

Ripken, H., 1989. Ertrag und Aufwand im Forstbetrieb — Stand und Tendenzen. Der Forst- und Holzwirt 44 (3), S51–S57.

Ripken, H., Spellmann, H., 1980. Modell-Berechnungen der Reinerträge der wichtigsten Baumarten sowie der gesamten Holzproduktion in den Niedersächsischen Landesforsten. Der Forst- und Holzwirt 35 (8), S153–S165.

Schreyer, G., 1986. Umtriebszeiten für Fichte und Kiefer aus der Sicht der Bayerischen Staatsforstverwaltung. Allgemeine Forst Zeitung 21, S517–S519.

Spiecker, H., Mielikäinen, K., Köhl, M., Skovsgaard, J.P., 1996 (Eds.). Growth Trends in European Forests — Studies from 12 Countries. European Forest Institute Research Report No. 5. Berlin: Springer Verlag.

Statistisches Bundesamt Wiesbaden, 1966. Forsterhebung 1961. Heft 2 Betriebsstruktur. Fachserie B, Land- und Forstwirtschaft, Fischerei. Stuttgart und Mainz: W. Kohlhammer Verlag.

Statistisches Reichsamt, 1930. Die Ergebnisse der forstwirtschaftlichen Erhebung im Jahre 1927. In: Statistik des Deutschen Reiches. Band 386. Berlin: Verlag von Reimar Hobbing.

Statistisches Reichsamt, 1943. Statistik der deutschen Forstwirtschaft (Forsterhebung 1937). In: Statistik des Deutschen Reichs. Band 592. Berlin: Verlag für Sozialpolitik, Wirtschaft und Statistik, Paul Schmidt.

Weidenbach, P., 1988. Grundsätze der Landesforstverwaltung Baden-Württemberg. Allgemeine Forst Zeitung 51, S1405–S1408.